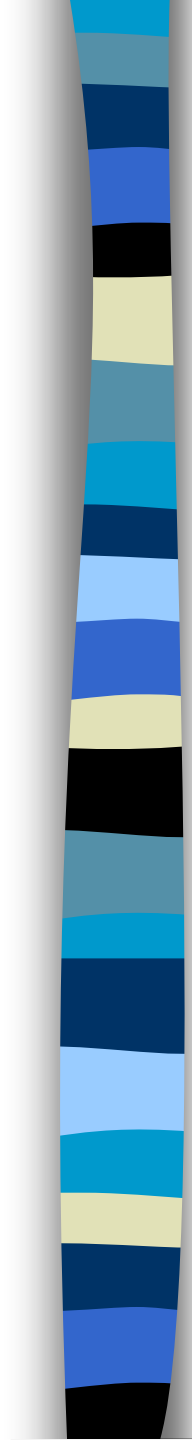


5.2 Protecting the data within database – Database Privileges: System privileges and Object privileges, Granting and Revoking Privileges: Grant and Revoke command.



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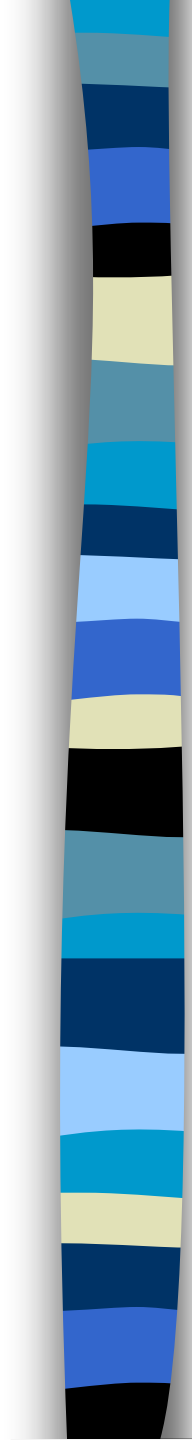


Database Privileges

- A **privilege** is a right to execute a particular type of SQL statement or to access another user's object. Some examples of privileges include the right to
 - connect to the database (create a session)
 - create a table
 - select rows from another user's table
 - execute another user's stored procedure
- We grant privileges to users so these users can accomplish tasks required for their job.
- Excessive granting of unnecessary privileges can compromise security.

Database Privileges

- A user can receive a privilege in two different ways:
 - We can grant privileges to users explicitly.
 - For example, we can explicitly grant the privilege to insert records into the EMP table to the user SCOTT.
 - We can also grant privileges to a role (a named group of privileges), and then grant the role to one or more users.
 - For example, we can grant the privileges to select, insert, update, and delete records from the EMP table to the role named CLERK, which in turn we can grant to the users SCOTT and STUDENT.
- Because roles allow for easier and better management of privileges, we should normally grant privileges to roles and not to specific users.
- There are two distinct categories of privileges:
 - system privileges
 - schema object privileges

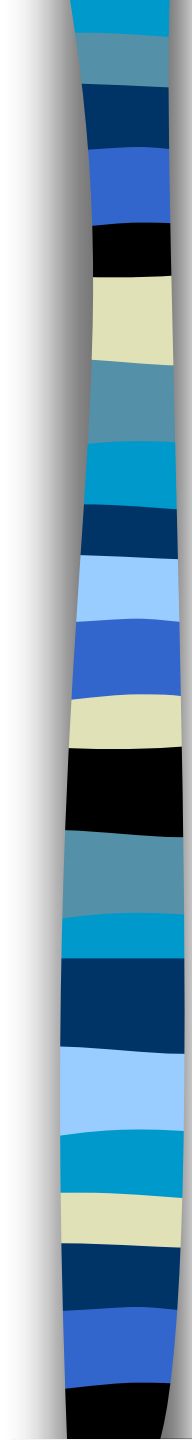


System Privileges

- System privileges are privileges given to users to allow them to perform certain functions that deal with *managing the database and the server* .
- This allows the user to CREATE, ALTER, or DROP database objects.

Examples of Oracle system privileges

- CREATE USER. The CREATE USER permission, when granted to a database user, allows that database user to create new users in the database.
- CREATE TABLE. The CREATE TABLE permission, when granted to a database user, allows that database user to create tables in their own schema. This type of privilege is also available for other object types – like stored procedures and indexes.
- CREATE SESSION. The CREATE SESSION permission, when granted to a database user, allows that database user to connect to the database.



Object Privileges

- Object privileges are privileges given to users so that they can perform certain actions upon certain database objects – where database objects are things like tables, stored procedures, indexes, etc.
- This allows the user to EXECUTE, SELECT, INSERT, UPDATE, or DELETE data from database objects to which the privileges apply.
- For example, granting a particular database user the right to DELETE and/or SELECT from a particular table.
- This is done using the GRANT clause.

SQL GRANT Command

SQL GRANT is a command used to provide access or privileges on the database objects to the users.

The Syntax for the GRANT command is:

**GRANT privilege_name ON object_name
TO {user_name |PUBLIC |role_name}
[WITH GRANT OPTION];**

privilege_name is the access right or privilege granted to the user. Some of the access rights are ALL, EXECUTE, and SELECT.

object_name is the name of a database object like TABLE, VIEW, STORED PROC and SEQUENCE.

user_name is the name of the user to whom an access right is being granted.

PUBLIC is used to grant access rights to all users.

ROLES are a set of privileges grouped together.

WITH GRANT OPTION - allows a user to grant access rights to other users.

GRANTING ROLES

Typically, we want to assign privileges to the user through attaching the account to various roles, starting with the CONNECT role:

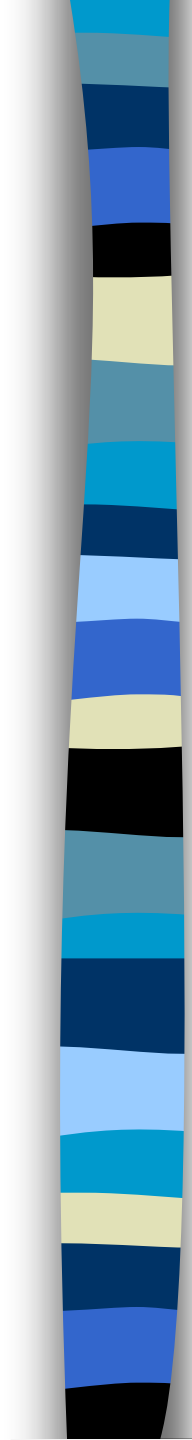
```
SQL>GRANT CONNECT TO STUDENT;
```

Grant succeeded.

```
SQL>GRANT CONNECT, RESOURCE, DBA, TO STUDENT;
```

Grant succeeded

Adding the RESOURCE role (allowing the user to create named types for custom schemas) or even the DBA role, which allows the user to not only create custom named types but alter and destroy them as well.



GRANTING PRIVILEGES

Provide new user with CREATE SESSION & TABLESPACE privilege

SQL>GRANT CREATE SESSION TO STUDENT;

Grant succeeded.

SQL>GRANT UNLIMITED TABLESPACE TO STUDENT;

Grant succeeded

SQL> GRANT SELECT ON employee TO user1;

Grant succeeded

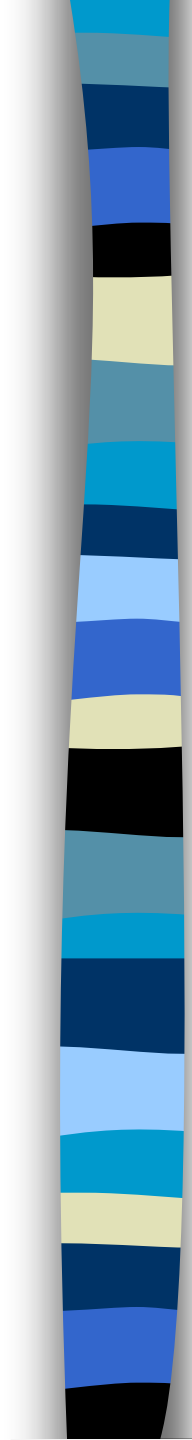


Table Privileges

- For example, if we want our student user to have the ability to perform SELECT, UPDATE, INSERT, and DELETE capabilities on the books table, we might execute the following GRANT statement:
- GRANT SELECT, INSERT, UPDATE, DELETE ON books TO student;

SQL REVOKE Command

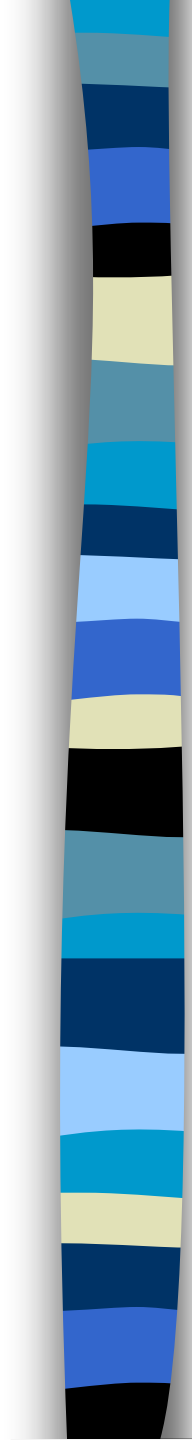
- The REVOKE command removes user access rights or privileges to the database objects.
- The Syntax for the REVOKE command is:

**REVOKE privilege_name
ON object_name
FROM {user_name |PUBLIC |role_name}**

SQL REVOKE Command - Example

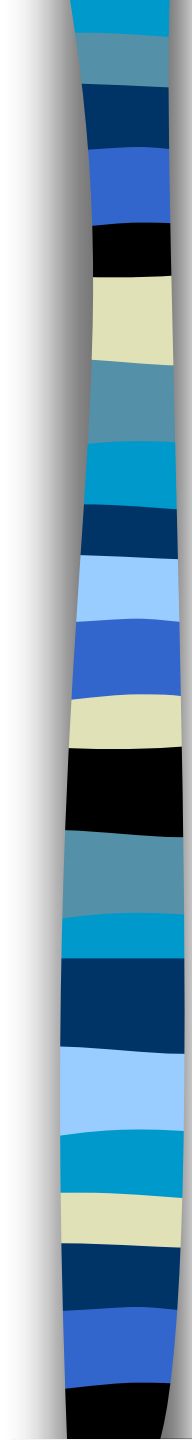
REVOKE SELECT ON employee FROM user1;

- This command will REVOKE a SELECT privilege on employee table from user1.
- When we REVOKE SELECT privilege on a table from a user, the user will not be able to SELECT data from that table anymore.
- However, if the user has received SELECT privileges on that table from more than one users, he/she can SELECT from that table until everyone who granted the permission revokes it.



Quick quiz

- The _____ is the gateway to the database.
- The _____ clause tells Oracle11g/12c how to authenticate a user account.
 - a. PASSWORD EXPIRE
 - b. IDENTIFIED
 - c. ACCOUNT
 - d. QUOTA



Quick Quiz

- SQL provides a command called _____ that removes a user account from the database
- When a user logs on to the database through the machine where the database is located, the database is called a _____.
 - a. local database
 - b. remote database
 - c. fixed database
 - d. database server